Report: Financial Transactions (Homework 3)

1. Modeling Decisions

The core objective of this data warehouse model is to capture and structure the trading activity recorded in the 2024 account statement. The architecture follows a star schema design, consisting of a central normalised fact table surrounded by four denormalised dimension tables. Each dimension is uniquely identified by an integer surrogate key generated during the ETL process.

To ensure maximum analytical granularity, the grain of the model is defined at the atomic event level; a single row within **Fact_Transactions** represents exactly one transaction from the source file.

The structural components of the schema are organised as follows:

- **Fact_Transactions:** Holds the primary surrogate key (transaction_sk), four foreign keys linking to the dimensions, a degenerate attribute (id_transaction), and an additive measure (units) designed for aggregation via SUM. Overall transaction volumes are derived using standard row counts (COUNT(*)).
- **Dim_Time:** Configured at the day level to support a chronological hierarchy (Day → Month → Quarter → Year), enriched with day_of_week as an additional descriptive attribute.
- **Dim_Symbol:** Captures asset classifications through a clear business hierarchy (Symbol → Industry → Sector).
- **Dim_Geography:** Models spatial relationships using ISO standards, organised from the most granular level upward (Country → Sub-region → Region).
- **Dim_TransactionType:** A flat dimension that categorises transaction types (such as BUY and SELL) without an internal hierarchy.

To eliminate data redundancy and preserve the integrity of the star schema, descriptive attributes are strictly isolated. For instance, the company's country is stored exclusively within **Dim_Geography**. The relationship between a specific ticker symbol and its corresponding country is dynamically resolved during the ETL load phase by performing a lookup against the master symbols dataset.

2. Data Quality Issues

An extensive profiling of the raw dataset revealed several structural and referential data quality issues, which were systematically addressed before loading the data warehouse:

- **Structural Anomalies:** The source file had some formatting problems: a number of empty rows and an extra empty column caused by a stray semicolon at the end of each line. Both were removed during cleaning, almost 464
- **Non-Unique Identifiers:** The source field IDTransaction was found to be unreliable for primary key: across the 2,069 transactions in the fact table there are only 1,124

distinct ID values, with 873 of them shared by more than one transaction. A synthetic surrogate key (transaction_sk) was implemented as the fact table's primary key.

- **Referential Integrity Breaks:** Out of 2,281 valid transactions, 212 records referenced 18 stock symbols entirely missing from the master symbols dataset. Because these orphan rows could not be accurately mapped to our dimensions, they were excluded from the fact table to maintain strict referential integrity, resulting in a final fact table of 2,069 rows.
- **Country Name Standardisation:** Discrepancies in country nomenclature were resolved by mapping non-standard names to their official ISO equivalents (specifically transforming "Turkey" to "Türkiye" and "Taiwan" to "Taiwan, Province of China").
- **Handling Missing Geographical Data:** The source entry for "Taiwan, Province of China" in country.csv lacked region and sub-region attributes. To prevent these transactions from being dropped during macro-level regional analysis, a manual patch was applied to correctly assign them to the "Asia-Eastern Asia" regional classification, aligning with the methodology demonstrated in class.
- **Transaction Scope Isolation:** Beyond standard trading operations, the dataset included 110 dividend-related rows (DIVIDENT). While these are maintained within **Dim_TransactionType** to ensure a comprehensive audit trail, they are explicitly filtered out during downstream analytical queries to avoid distorting the core BUY/SELL trading metrics.

3. Analytical Results

The finalised dataset was utilised to address key analytical objectives, specifically questions 3, 4, 5, 7, and 11, to validate the model's hierarchies. The initial findings reveal a highly concentrated operational trend:

- **Q3 - Quarters by number of transactions (BUY+SELL):** Market activity is heavily front-loaded into the first half of the year. The first quarter (Q1) exhibits overwhelming dominance with 968 transactions. Volume then contracts significantly in Q2 to 522 transactions before stabilising at a much lower baseline during the latter half of the year, with Q3 and Q4 recording 242 and 241 transactions, respectively.

- **Q4 - Top countries by SELL:** the United States dominates (389), followed by the UK (130) and China (112).

- **Q5 - Regions by units bought:** the Americas lead (37,026 units), ahead of Europe (22,528) and Asia (11,537).

- **Q7 - Most traded symbols:** technology names dominate, led by ARM (100), AMD (97) and TSM (77).

- **Q11 - Sub-regions in Q4:** Northern America is first (124 transactions), followed by Eastern Asia (39).

4. Streamlit Dashboard

An Interactive **Streamlit Dashboard**. The dashboard is a Streamlit application structured as shown in class (Home.py as entry point, a pages/ folder for the views, a utils/ module with a cached data loader, and a data/ folder for the model CSVs). The Time Analysis page provides a date-range filter (default 01/01/2024 – 31/12/2024) and four charts:

- A line chart showing the total number of transactions (BUY + SELL) over time.

- A bar chart showing the top 3 traded symbols by transaction count.

- A bar chart showing the top 5 sectors by transaction count.

- A bar chart showing the top 5 industries by transaction count.

all driven by a single filtered dataframe, so they react together to the selected range.